In A.C operation, both field and armature currents change their polarities, at the same time resulting in unidirectional torque.

APPLICATION

Universal motor is used in:

1. Vacuum cleaners
2. Food mixer
3. Portable drills and
4. Domestic sewing machines

8.5 TYPES OF THREE PHASE MOTORS

Three phase induction motors are widely used in industrial and domestic drives. The rotor of induction motor receives the power from the stator by means of induction. Most of the operating principles resembles to the working of transformer only. So, it can also be treated as a rotating transformer. That is, primary winding is stationary part and secondary winding is rotating part.

According to the rotor construction, three phase induction motors are classified as

i) 3-phase squirrel cage induction motor
ii) 3-phase slip-ring induction motor

8.5.1 Principle of Operation

Three phase supply is given to the stator winding. Due to this, current flows through the stator winding and produces a rotating magnetic field in the space between stator and rotor. This magnetic field rotates at synchronous speed given by $N_s = \frac{120f}{p}$.

Where N_s = Synchronous speed

f = Supply frequency

p = Number of poles for which the stator is wound

This rotating magnetic field cuts the rotor conductors, an emf is induced in the rotor. This induced emf produces current and this current produces a rotor field.

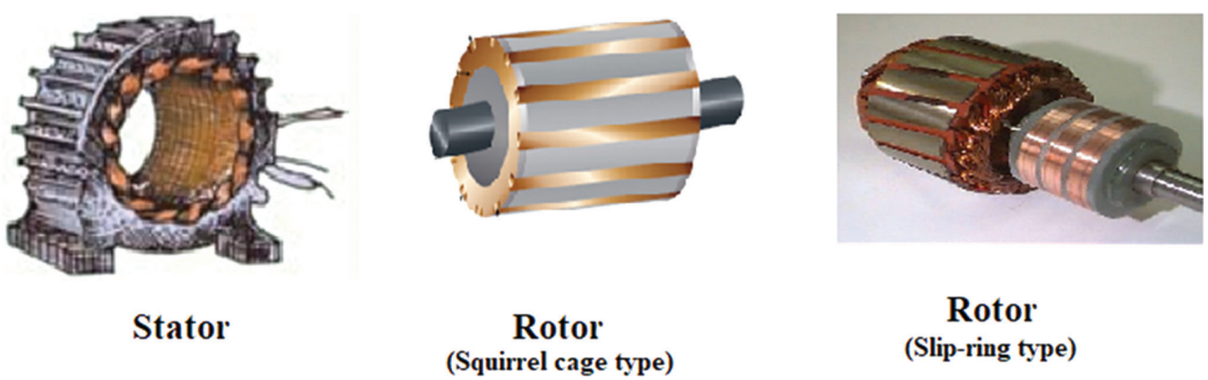


Fig 8.10 *Stator and Rotor of 3 Phase Induction Motor*

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